
Student Management System

Java Project for Object Oriented Programming

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Object Oriented Programming

Presented by
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A desktop application for managing student records with a structured, scalable architecture.

Key features and technologies:

- Manages student records stored in a MySQL database
- GUI developed with Java Swing; database connectivity via JDBC
- Demonstrates CRUD operations, event handling, and OOP principles
- Scalable design allowing future feature additions

features over time.

1. Project Overview

The Student Management System is a desktop application that helps an administrator manage student records. Instead of storing information in files or on paper, all data is stored in a **MySQL database**. The application provides a graphical user interface (GUI) built using **Java Swing**, while **JDBC** is used to connect Java to the database.

2. Objectives

The main objectives are to:

- Store student information in a database.
- Perform CRUD (Create, Read, Update, Delete) operations.
- Search for students quickly.
- Display records in a table.

Project Objectives

- 1 Secure Storage**
Store student information securely in a MySQL database, ensuring data integrity and reliable access through structured records.
- 2 CRUD Operations**
Perform Create, Read, Update, and Delete operations efficiently, enabling full lifecycle management of student records via JDBC.
- 3 Search & Display**
Enable quick student search functionality and display records in organized tables for clear and efficient data retrieval.
- 4 Intuitive GUI**
Provide a user-friendly graphical interface built with Java Swing, minimizing manual record keeping and reducing input errors.

Main interface functionalities available post-login.

Available dashboard controls:

Add Student — register new student records

Update Student — modify existing information

Delete Student — remove student entries

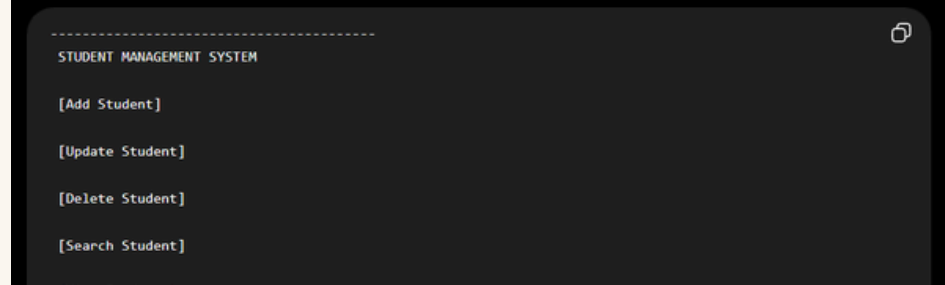
Search Student — locate records quickly

View All Students — display full records in table

Logout — securely exit the system

- Add Student
- Update Student
- Delete Student
- Search Student
- View All Students
- Logout

Example:



Robust error handling and a roadmap for future growth

The system handles empty fields, invalid or duplicate IDs, login failures, database connection issues, and SQL exceptions to ensure stability.

Planned enhancements include photo upload, attendance tracking, grade management, course registration, fee tracking, report generation, data export, role-based logins, password encryption, and dark mode support.

13. Future Enhancements

You can extend the project with:

- Student photo upload
- Attendance management
- Marks and grades
- Course registration
- Fee payment tracking
- PDF report generation
- CSV/Excel export
- Role-based login (Admin/Teacher)

Thank You

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